

Q1 (3)

$$\text{Normal vector of required plane is } \vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & -2 \\ 2 & -5 & -1 \end{vmatrix} = -11\hat{i} - \hat{j} - 17\hat{k}$$

$$\therefore 11(x-1) + (y-2) + 17(z+3) = 0$$

$$11x + y + 17z + 38 = 0$$

Q2 (1)

$$\frac{x-3}{1} = \frac{y-4}{2} = \frac{z-5}{2} = \lambda$$

$$\Rightarrow x = \lambda + 3, y = 2\lambda + 4, z = 2\lambda + 5$$

Which lines on given plane hence

$$\Rightarrow \lambda + 3 + 2\lambda + 4 + 2\lambda + 5 = 17$$

$$\Rightarrow \lambda = \frac{5}{3} = 1$$

Hence, point of intersection is Q(4, 6, 7)

 $\therefore$ Required distance = PQ

$$= \sqrt{9 + 25 + 4}$$

$$= \sqrt{38}$$

Q3 (2)

$$P(a, 6, 9), Q(20, b, -a-9)$$

$$\text{mid point of PQ} = \left(\frac{a+20}{2}, \frac{b+6}{2}, -\frac{a}{2} \right)$$

lie on line

$$\frac{\frac{a+20}{2}-3}{7} = \frac{\frac{b+6}{2}-2}{5} = \frac{-\frac{a}{2}-1}{-9}$$

$$\frac{a+20-6}{14} = \frac{b+6-4}{10} = \frac{-a-2}{-18}$$

$$\frac{a+14}{14} = \frac{a+2}{18}$$

$$18a + 252 = 14a + 28$$

Hints and Solutions

MathonGo

$$4a = -224$$

$$a = -56$$

$$\frac{b+2}{10} = \frac{a+2}{18}$$

$$\frac{b+2}{10} = \frac{-54}{18}$$

$$\frac{b+2}{10} = -3 \Rightarrow b = -32$$

$$|a + b| = |-56 - 32| = 88$$

Q4 (2)

Plane passing through intersection of plane is $\{\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) - 1\} + \lambda \{\vec{r} \cdot (\hat{i} - 2\hat{j}) + 2\} = 0$

Passes through $\hat{i} + 2\hat{k}$, we get

$$(3 - 1) + \lambda(1 + 2) = 0 \Rightarrow \lambda = -\frac{2}{3}$$

Hence, equation of plane is $3\{\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) - 1\} - 2\{\vec{r} \cdot (\hat{i} - 2\hat{j}) + 2\} = 0$

$$\Rightarrow \vec{r} \cdot (\hat{i} + 7\hat{j} + 3\hat{k}) = 7$$

Q5 (1)

$$\frac{x-\lambda}{1} = \frac{y-\frac{1}{2}}{\frac{1}{2}} = \frac{z}{-\frac{1}{2}}$$

$$\frac{x-\lambda}{2} = \frac{y-\frac{1}{2}}{1} = \frac{z}{-1} \dots\dots (1) \quad \text{Point on line} = (\lambda, \frac{1}{2}, 0)$$

$$\frac{x}{1} = \frac{y+2\lambda}{1} = \frac{z-\lambda}{1} \quad \text{Point on line} = (0, -2\lambda, \lambda)$$

$$\text{Distance between skew lines} = \frac{[\vec{a}_2 - \vec{a}_1 \quad \vec{b}_1 \quad \vec{b}_2]}{|\vec{b}_1 \times \vec{b}_2|}$$

$$\frac{\begin{vmatrix} \lambda & \frac{1}{2} + 2\lambda & -\lambda \\ 2 & 1 & -1 \\ 1 & 1 & 1 \end{vmatrix}}{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 1 & 1 & 1 \end{vmatrix}}$$

$$= \frac{|-5\lambda - \frac{3}{2}|}{\sqrt{14}} = \frac{\sqrt{7}}{2\sqrt{2}} \quad (\text{given})$$

$$= |10\lambda + 3| = 7 \Rightarrow \lambda = -1$$

$$\Rightarrow |\lambda| = 1$$

Q6 (3)

Hints and Solutions

MathonGo

$$l^2 + m^2 + n^2 = 1$$

$$\therefore 2n^2 = 1 \Rightarrow n = \pm \frac{1}{\sqrt{2}}$$

$$\therefore l^2 + m^2 = \frac{1}{2} \text{ \& } l + m = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \frac{1}{2} - 2lm = \frac{1}{2}$$

$$\Rightarrow 1 - 4lm = 1 \text{ or } m = 0$$

$$\therefore l = 0, m = \frac{1}{\sqrt{2}} \text{ or } l = \frac{1}{\sqrt{2}}$$

$$< 0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} > \text{ or } < \frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} >$$

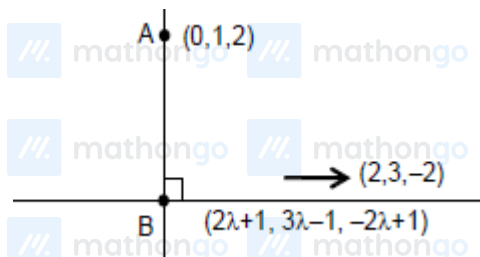
$$\therefore \cos \alpha = 0 + 0 + \frac{1}{2} = \frac{1}{2}$$

$$\therefore \sin^4 \alpha + \cos^4 \alpha = 1 - \frac{1}{2} \sin^2(2\alpha) = 1 - \frac{1}{2}, \frac{3}{4} = \frac{5}{8}$$

Q7 (1)

$$\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{-2} = \lambda$$

Any point on this line $(2\lambda + 1, 3\lambda - 1, -2\lambda + 1)$



Direction ratio of given line $(2, 3, -2)$

Direction ratio of line to be found $(2\lambda + 1, 3\lambda - 2, -2\lambda - 1) \therefore \vec{d}_1 \cdot \vec{d}_2 = 0$

$$\lambda = 2/17$$

Direction ratio of line $(21, -28, -21) \equiv (3, -4, -3) \equiv (-3, 4, 3)$

Q8 (3)

$A(1, 2, 3), B(2, 3, 1), C(2, 4, 2), O(0, 0, 0)$

Equation of plane passing through A, B, C will be

Hints and Solutions

MathonGo

$$\begin{vmatrix} x-1 & y-2 & z-3 \\ 2-1 & 3-2 & 1-3 \\ 2-1 & 4-2 & 2-3 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x-1 & y-2 & z-3 \\ 1 & 1 & -2 \\ 1 & 2 & -1 \end{vmatrix} = 0$$

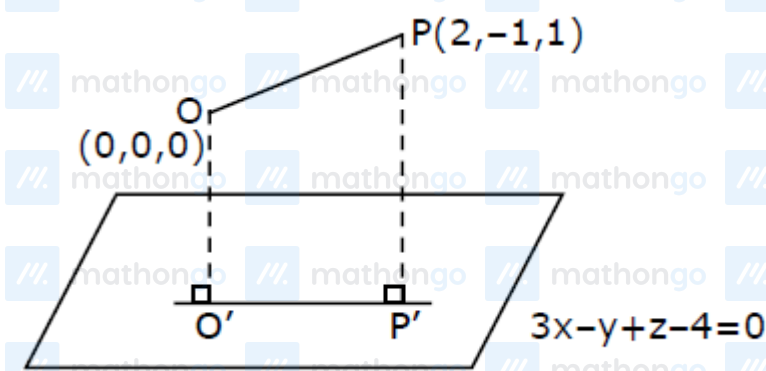
$$\Rightarrow (x-1)(-1+4) - (y-2)(-1+2) + (z-3)(2-1) = 0$$

$$\Rightarrow (x-1)(3) - (y-2)(1) + (z-3)(1) = 0$$

$$\Rightarrow 3x - 3 - y + 2 + z - 3 = 0$$

$$\Rightarrow 3x - y + z - 4 = 0, \text{ is the required plane.}$$

Now, given $O(0,0,0)$ & $P(2,-1,1)$



Plane is $3x - y + z - 4 = 0$

O' & P' are foot of perpendiculars.

for O'

$$\frac{x-0}{3} = \frac{y-0}{-1} = \frac{z-0}{1} = \frac{-(0-0+0-4)}{9+1+1}$$

$$\frac{x}{3} = \frac{y}{-1} = \frac{z}{1} = \frac{4}{11}$$

$$\Rightarrow O' \left(\frac{12}{11}, \frac{-4}{11}, \frac{4}{11} \right)$$

for P'

$$\frac{x-2}{3} = \frac{y+1}{-1} = \frac{z-1}{1} = \frac{-(3(2)-(-1)+1-4)}{9+1+1}$$

$$\frac{x-2}{3} = \frac{y+1}{-1} = \frac{z-1}{1} = \left(\frac{-4}{11} \right)$$

$$P' \left(\frac{-12}{11} + 2, \frac{4}{11} - 1, \frac{-4}{11} + 1 \right)$$

$$\Rightarrow P' \left(\frac{10}{11}, \frac{-7}{11}, \frac{7}{11} \right)$$

$$O'P' = \sqrt{\left(\frac{10}{11} - \frac{12}{11} \right)^2 + \left(\frac{-7}{11} + \frac{4}{11} \right)^2 + \left(\frac{7}{11} - \frac{4}{11} \right)^2}$$

Hints and Solutions

MathonGo

$$\Rightarrow OP = \frac{1}{11} \sqrt{4+9+9}$$

$$\Rightarrow OP' = \frac{\sqrt{22}}{11}$$

$$\Rightarrow OP' = \frac{\sqrt{2 \times 11}}{11}$$

$$\Rightarrow OP = \sqrt{\frac{2}{11}}$$

Q9 (44)

$$\ell_1 : \vec{r} = (3+t)\hat{i} + (-1+2t)\hat{j} + (4+2t)\hat{k}$$

$$\ell_1 : \frac{x-3}{1} = \frac{y+1}{2} = \frac{z-4}{2} \Rightarrow \text{D. R. of } \ell_1 = 1, 2, 2$$

$$\ell_2 : \vec{r} = (3+2s)\hat{i} + (3+2s)\hat{j} + (2+s)\hat{k}$$

$$\ell_2 : \frac{x-3}{2} = \frac{y-3}{2} = \frac{z-2}{1} \Rightarrow \text{D. R. of } \ell_2 = 2, 2, 1$$

D. R. of ℓ is $\perp$ to ℓ_1 & ℓ_2

$$\therefore \text{D.R. of } \ell \parallel (\ell_1 \times \ell_2) \Rightarrow \langle -2, 3, -2 \rangle$$

$$\therefore \text{Equation of } \ell : \frac{x}{2} = \frac{y}{-3} = \frac{z}{2}$$

Solving ℓ & ℓ_1

$$(2\lambda, -3\lambda, 2\lambda) = (\mu + 3, 2\mu - 1, 2\mu + \mu)$$

$$\Rightarrow 2\lambda = \mu + 3$$

$$-3\lambda = 2\mu - 1$$

$$2\lambda = 2\mu + 4$$

$$\mu = -1$$

$$\mu = -1$$

$P(2, -3, 2)$ { intersection point }

Let, $Q(2v+3, 2v+3, v+2)$ be point on ℓ_2

$$\text{Now, } PQ = \sqrt{(2v+3-2)^2 + (2v+3+3)^2 + (v+2-2)^2} = \sqrt{17}$$

$$\Rightarrow (2v+1)^2 + (2v+6)^2 + (v)^2 = 17$$

$$\Rightarrow 9v^2 + 28v + 36 + 1 - 17 = 0$$

$$\Rightarrow 9v^2 + 28v + 20 = 0$$

$$\Rightarrow 9v^2 + 18v + 10v + 20 = 0$$

$$\Rightarrow (9v+10)(v+2) = 0$$

Hints and Solutions

MathonGo

$$\Rightarrow v = -2(\text{rejected}), -\frac{10}{9}(\text{accepted})$$

$$Q(3 - \frac{20}{9}, 3 - \frac{20}{9}, 2 - \frac{10}{9})$$

$$(\frac{7}{9}, \frac{7}{9}, \frac{8}{9})$$

$$\therefore 18(a+b+c)$$

$$= 18(\frac{7}{9} + \frac{7}{9} + \frac{8}{9})$$

$$= 44$$

Q10 (1)

$$P_1 = x + 5y + 7z = 3$$

$$P_2 = x - 3y - z = 5$$

$$P_3 = x + 5y + 7z = 5/2$$

$$\Rightarrow P_1 \parallel P_3$$

Q11 (4)

$$\text{Let } P(1, 5, 35), Q(7, 5, 5), R(1, \lambda, 7), S(2\lambda, 1, 2)$$

$$[\vec{PQ} \vec{PR} \vec{PS}] = 0$$

$$\begin{vmatrix} 6 & 0 & -30 \\ 0 & \lambda - 5 & -28 \\ 2\lambda - 1 & -4 & -33 \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & 0 & -5 \\ 0 & \lambda - 5 & -28 \\ 2\lambda - 1 & -4 & -33 \end{vmatrix} = 0$$

$$\{-33\lambda + 165 - 112\} + 5(\lambda - 5)(2\lambda - 1) = 0$$

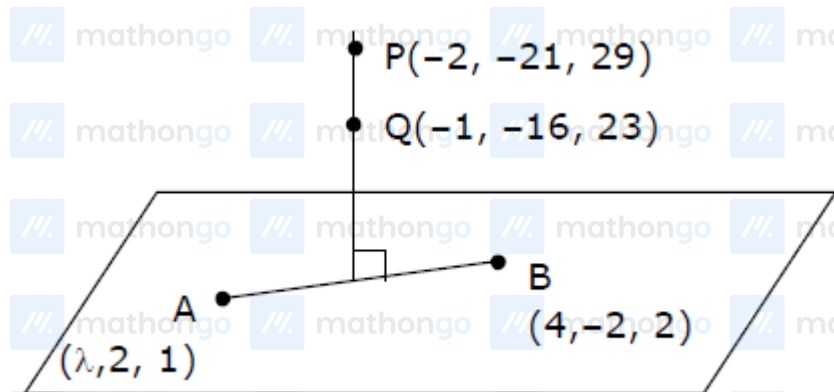
$$53 - 33\lambda + 5\{2\lambda^2 - 11\lambda + 5\} = 0$$

$$16\lambda^2 - 88\lambda + 78 = 0$$

$$5\lambda^2 - 44\lambda + 39 = 0 \begin{matrix} \lambda_1 \\ \lambda_2 \end{matrix}$$

$$\Rightarrow \lambda_1 + \lambda_2 = 44/5$$

Q12 (8)



$$\vec{AB} \perp \vec{PQ}$$

$$[(4-\lambda)\hat{i} - 4\hat{j} + \hat{k}] \cdot [\hat{i} + 5\hat{j} - 6\hat{k}] = 0$$

$$4 - \lambda - 20 - 6 = 0$$

$$\lambda = -22$$

$$\text{Now, } \frac{\lambda}{11} = -2$$

$$\Rightarrow \left(\frac{\lambda}{11}\right)^2 - \frac{4\lambda}{11} - 4$$

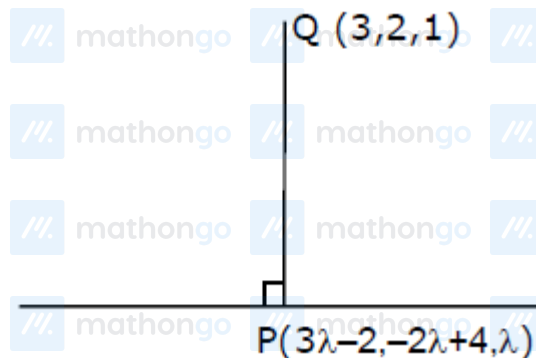
$$\Rightarrow 4 + 8 - 4 = 8$$

Q13 (4)

$$\text{Dir's of line } \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{vmatrix} = 3\hat{i} - 2\hat{j} + \hat{k}$$

$$\text{Dir's : } -(3, -2, 1)$$

$$\text{Points on the line } (-2, 4, 0) \text{ Equation of the line } \frac{x+2}{3} = \frac{y-4}{-2} = \frac{z}{1} = \lambda$$



Hints and Solutions

MathonGo

Dr's of PQ; $3\lambda - 5, -2\lambda + 2, \lambda - 1$

Dr's of y lines are $(3, -2, 1)$ Since $PQ \perp$ line $3(3\lambda - 5) - 2(-2\lambda + 2) + 1(\lambda - 1) = 0$

$$\lambda = \frac{10}{7}$$

$$P\left(\frac{16}{7}, \frac{8}{7}, \frac{10}{7}\right)$$

$$21(\alpha + \beta + \gamma) = 21\left(\frac{34}{7}\right) = 102$$

Q14 (I)

Image of $(1, 3, 5)$ in the plane $4x - 5y + 2z = 8$ is (α, β, γ)

$$\Rightarrow \frac{\alpha-1}{4} = \frac{\beta-3}{-5} = \frac{\gamma-5}{2} = -2 \frac{(4(1)-5(3)+2(5)-8)}{4^2+5^2+2^2} = \frac{2}{5}$$

$$\therefore \alpha = 1 + 4\left(\frac{2}{5}\right) = \frac{13}{5}$$

$$\beta = 3 - 5\left(\frac{2}{5}\right) = 1 = \frac{5}{5}$$

$$\gamma = 5 + 2\left(\frac{2}{5}\right) = \frac{29}{5}$$

$$\text{Thus, } 5(\alpha + \beta + \gamma) = 5\left(\frac{13}{5} + \frac{5}{5} + \frac{29}{5}\right) = 47$$

Questions with Answer Keys

MathonGo

Q1: 24 Feb (Shift 1) - Single Correct

The equation of the plane passing through the point $(1, 2, -3)$ and perpendicular to the planes $3x + y - 2z = 5$ and $2x - 5y - z = 7$, is:

(1) $3x - 10y - 2z + 11 = 0$

(2) $6x - 5y - 2z - 2 = 0$

(3) $11x + y + 17z + 38 = 0$

(4) $6x - 5y + 2z + 10 = 0$

Q2: 24 Feb (Shift 1) - Single Correct

The distance of the point $(1, 1, 9)$ from the point of intersection of the line $\frac{x-3}{1} = \frac{y-4}{2} = \frac{z-5}{2}$ and the plane $x + y + z = 17$ is:

(1) $\sqrt{38}$

(2) $19\sqrt{2}$

(3) $2\sqrt{19}$

(4) 38

Q3: 24 Feb (Shift 2) - Single Correct

Let $a, b \in \mathbb{R}$. If the mirror image of the point $P(a, 6, 9)$ with respect to the line

$\frac{x-3}{7} = \frac{y-2}{5} = \frac{z-1}{-9}$ is $(20, b, -a-9)$, then $|a+b|$ is equal to:

(1) 86

(2) 88

(3) 84

(4) 90

Q4: 24 Feb (Shift 2) - Single Correct

The vector equation of the plane passing through the intersection of the planes $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 1$ and $\vec{r} \cdot (\hat{i} - 2\hat{j}) = -2$, and the point $(1, 0, 2)$ is:

Questions with Answer Keys

MathonGo

$$(1) \vec{r} \cdot (\hat{i} - 7\hat{j} + 3\hat{k}) = \frac{7}{3}$$

$$(2) \vec{r} \cdot (\hat{i} + 7\hat{j} + 3\hat{k}) = 7$$

$$(3) \vec{r} \cdot (3\hat{i} + 7\hat{j} + 3\hat{k}) = 7$$

$$(4) \vec{r} \cdot (\hat{i} + 7\hat{j} + 3\hat{k}) = \frac{7}{3}$$

Q5: 24 Feb (Shift 2) - Numerical

Let λ be an integer. If the shortest distance between the lines $x - \lambda = 2y - 1 = -2z$ and

$x = y + 2\lambda = z - \lambda$ is $\frac{\sqrt{7}}{2\sqrt{2}}$, then the value of $|\lambda|$ is

Q6: 25 Feb (Shift 1) - Single Correct

Let α be the angle between the lines whose direction cosines satisfy the equations $1 + m - n = 0$ and

$l^2 + m^2 - n^2 = 0$. Then the value of $\sin^4 \alpha + \cos^4 \alpha$ is :

$$(1) \frac{3}{4}$$

$$(2) \frac{1}{2}$$

$$(3) \frac{5}{8}$$

$$(4) \frac{3}{8}$$

Q7: 25 Feb (Shift 1) - Single Correct

The equation of the line through the point $(0, 1, 2)$ and perpendicular to the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{-2}$ is :

$$(1) \frac{x}{-3} = \frac{y-1}{4} = \frac{z-2}{3}$$

$$(2) \frac{x}{3} = \frac{y-1}{4} = \frac{z-2}{3}$$

$$(3) \frac{x}{3} = \frac{y-1}{-4} = \frac{z-2}{3}$$

$$(4) \frac{x}{3} = \frac{y-1}{4} = \frac{z-2}{-3}$$

Q8: 25 Feb (Shift 2) - Single Correct

A plane passes through the points $A(1, 2, 3)$, $B(2, 3, 1)$ and $C(2, 4, 2)$. If O is the origin and P is $(2, -1, 1)$, then the projection of $\vec{OP}$ on this plane is of length:

Questions with Answer Keys

MathonGo

(1) $\sqrt{\frac{2}{5}}$

(2) $\sqrt{\frac{2}{3}}$

(3) $\sqrt{\frac{2}{11}}$

(4) $\sqrt{\frac{2}{7}}$

Q9: 25 Feb (Shift 2) - Numerical

A line ' ℓ ' passing through origin is perpendicular to the lines

$$\ell_1 : \vec{r} = (3 + t)\hat{i} + (-1 + 2t)\hat{j} + (4 + 2t)\hat{k}$$

$$\ell_2 : \vec{r} = (3 + 2s)\hat{i} + (3 + 2s)\hat{j} + (2 + s)\hat{k}$$

If the co-ordinates of the point in the first octant on ' ℓ_2 ' at the distance of $\sqrt{17}$ from the point of intersection of ' ℓ ' and ' ℓ_1 ' are (a, b, c) , then $18(a + b + c)$ is equal to

Q10: 26 Feb (Shift 1) - Single Correct

Consider the three planes

$$P_1 : 3x + 15y + 21z = 9$$

$$P_2 : x - 3y - z = 5, \text{ and}$$

$$P_3 : 2x + 10y + 14z = 5$$

Then, which one of the following is true?

(1) P_1 and P_3 are parallel.

(2) P_2 and P_3 are parallel.

(3) P_1 and P_2 are parallel.

(4) P_1, P_2 and P_3 all are parallel.

Q11: 26 Feb (Shift 1) - Single Correct

If $(1, 5, 35), (7, 5, 5), (1, \lambda, 7)$ and $(2\lambda, 1, 2)$ are coplanar, then the sum of all possible values of λ is:

(1) $-\frac{44}{5}$

Questions with Answer Keys

MathonGo

(2) $\frac{39}{5}$

(3) $-\frac{39}{5}$

(4) $\frac{44}{5}$

Q12: 26 Feb (Shift 1) - Numerical

Let $(\lambda, 2, 1)$ be a point on the plane which passes through the point $(4, -2, 2)$. If the plane is perpendicular to the line joining the point $(-2, -21, 29)$ and $(-1, -16, 23)$, then

$$\left(\frac{\lambda}{11}\right)^2 - \frac{4\lambda}{11} - 4$$
 is equal to

Q13: 26 Feb (Shift 2) - Single Correct

Let L be a line obtained from the intersection of two planes $x + 2y + z = 6$ and $y + 2z = 4$. If point

$P(\alpha, \beta, \gamma)$ is the foot of perpendicular from $(3, 2, 1)$ on L , then the value of $21(\alpha + \beta + \gamma)$ equals :

(1) 142

(2) 68

(3) 136

(4) 102

Q14: 26 Feb (Shift 2) - Single Correct

If the mirror image of the point $(1, 3, 5)$ with respect to the plane $4x - 5y + 2z = 8$ is (α, β, γ) , then $5(\alpha + \beta + \gamma)$ equals:

(1) 47

(2) 39

(3) 43

(4) 41

Answer Key**Q1 (3)****Q2 (1)****Q3 (2)****Q4 (2)****Q5 (1)****Q6 (3)****Q7 (1)****Q8 (3)****Q9 (44)****Q10 (1)****Q11 (4)****Q12 (8)****Q13 (4)****Q14 (1)**